

```
1 #roots in quadratic divison
2 import math
3 a=int(input("enter a:"))
4 b=int(input("\nenter b:"))
5 c=int(input("\nenter c:"))
6 d=b*b-4*a*c
7 if d>=0:
8     root1=(-b+math.sqrt(d))/(2*a)
9     root2=(-b-math.sqrt(d))/(2*a)
10    print("\nthe first root:",root1)
11    print("\nthe second root:",root2)
12 else:
13    print("\nthe imaginary roots:")
```

-1
-2
3

OUTPUT

```
enter a:
enter b:
enter c:
the first root: -3.0

the second root: 1.0
```

```
1 num=int(input("enter the number:"))
2 a=0
3 i=1
4 while(i<=num):
5     a=a+i
6     i+=1
7 print(a)
```

5

```
enter the number:15
```

```
1 n=input("enter the number:")
2 a=int(n)
3 b=0
4 while(a>0):
5     b=b*10+a%10
6     a//=10
7 if(int(n)==b):
8     print("\nthe given number is palindrome")
9 else:
10    print("\nthe given number is not a palindrome")
```

120

OUTPUT

```
enter the number:
the given number is not a palindrome
```

Factorial

FACTORIAL

Write a Python program to find the factorial of a number using a loop.

PROGRAM EDITOR

Python

Run

Save

```
1 num=int(input("enter the number:"))
2 i=1
3 a=1
4 while(i<=num):
5     a=a*i
6     i+=1
7 print("\nthe factorial",a)
```

OUTPUT

```
enter the number:
the factorial 120
```

INPUT

5

```
1 #reverse of the number
2 num=int(input("enter the number:"))
3 rev=0
4 a=1
5 while(num>0):
6     a=num%10
7     rev=rev*10+a
8     num//=10
9 print("\nthe reverse is",rev)
```

OUTPUT

```
enter the number:
the reverse is 321
```

```
1 #add from 1 to 20
2 n=int(input("enter the number:"))
3 i=1
4 while i<=n:
5     if(i%2!=0):
6         print(i,end=" ")
7     i+=1
```

20

```
enter the number:1 3 5 7 9 11 13 15 17 19
```

```
1 #fibonacci series
2 n=int(input("enter the number:"))
3 i=1
4 a=0
5 b=1
6 while(i<=n):
7     print(a,end=" ")
8     c=a+b
9     a=b
10    b=c
11    i+=1
12
```

OUTPUT

```
enter the number:0 1 1 2 3 5
```

```
1 #sum of even fibonacci series
2 n=int(input("enter the number:"))
3 i=1
4 a=0
5 b=1
6 d=0
7 while i<=n:
8     if a%2==0:
9         d+=a
10    c=a+b
11    a=b
12    b=c
13    i+=1
14 print(d)
```

```
enter the number:44
```

```
1 #sum of given number
2 num=int(input("enter the number:"))
3 b=0
4 while num>0:
5     b+=num%10
6     num//=10
7 print(b)
```

5

```
enter the number:5
```

```
1 #armstrong number
2 a=input("enter the number")
3 b=0
4 n=int(a)
5 while n>0:
6     b+=(n%10)**len(a)
7     n//=10
8 if int(a)==b:
9     print("\nit is armstrong number")
10 else:
11     print("\nIt is not a armstrong number")
```

15

OUTPUT

```
enter the number
It is not a armstrong number
```

PROGRAM EDITOR

Python



Run

Save

```
1 #1 to 10
2 i=1
3 while(i<=10):
4     print(i,end=" ")
5     i+=1
```

OUTPUT

```
1 2 3 4 5 6 7 8 9 10
```

```
1 #sum in range
2 num=int(input("enter the number:"))
3 sum=0
4 for i in range(0,num):
5     sum=sum+i
6 print("\nsum=",sum)
```

10

```
enter the number:sum= 45
```

```
1 #sum of already given
2 num={2,4,6,5,3,10}
3 sum=0
4 for i in num:
5     sum+=i
6 print("sum=",sum)
```

OUTPUT

```
sum= 24
```

```
1 #factorail of given number in range
2 num=int(input("enter the number:"))
3 fact=1
4 for i in range(1,num+1):
5     fact*=i
6 print("\nthe factorial =",fact)
```

5

OUTPUT

```
enter the number:the factorial = 120
```

PROGRAM EDITOR

Python



Run

Save

```
1 #to find factors
2 num=int(input("enter the number:"))
3 for i in range(1,num+1):
4     if(num%i==0):
5         print(i,end=" ")
6
```

INPUT

20

OUTPUT

```
enter the number:1
```

```
2
```

```
4
```

```
5
```

```
10
```

```
15
```

```
1 #PERFECT NUMBER
2 num=int(input("enter the number:"))
3 sum=0
4 for i in range(1,num):
5     if(num%i==0):
6         sum=sum+i
7 if sum==num:
8     print("\nit is perfect")
9 else:
10    print("\nit is not a perfect number")
```

6

```
enter the number:
it is perfect
```

```
1 #perfect squares
2 num=int(input("enter the number:"))
3 for i in range(1,num+1):
4     if i*i==num:
5         print("\nperfect number")
6         break
7 else:
8     print('\nnot a perfect number')
```

49

```
enter the number:perfect number
```

```
1 #check prime or not
2 num=int(input("enter the number"))
3 for i in range (2,num):
4     if num%i==0:
5         print("\nnot a prime number")
6         break
7 else:
8     print("\nit is a prime number")
```

OUTPUT

```
enter the numberit is a prime number
```

```
1 #prime within a range
2 for n in range(2,21):
3     for i in range(2,n):
4         if n%i==0:
5             break
6     else:
7         print(n,end=" ")
```

Your INPUT goes here!

```
2
3
5
7
11
```

```
1 #prime number next
2 num=int(input("enter the number"))
3 b=num+1
4 while True:
5     for i in range(2,b):
6         if b%i==0:
7             b+=1
8             break
9     else:
10         print("\nnext prime",b)
11         break
```

13

```
enter the numbernext prime 17
```